Linear systems and elimination

Optimal uniform row-deletion singular-value constant

Difficulty: challenging **Importance:** interesting to the community

Status: open in the checked literature

Problem statement

Fix $0 < \theta < 1$. For a real $m \times n$ matrix A with every row of Euclidean norm one, put

$$s_\theta(A) = \min_{S \subseteq \{1,\dots,m\},\ |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2 = 1} \|A_S x\|_2, \qquad M_{m,n}(\theta) = \sup_{\|a_i\|_2 = 1} \sqrt{\frac{n}{m}}\, s_\theta(A).$$

Here a_i is row i of A, and A_S contains exactly the rows indexed by S.

Determine the sharp asymptotic upper constant for $M_{m,n}(\theta)$ as $n \to \infty$ and $m/n \to \infty$. In particular, is it

$$c_\theta = \left(\frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2}\, dt \right)^{1/2}, \qquad \mathbb{P}(|G| \le a_\theta) = \theta, \quad G \sim N(0,1)?$$

Precisely, the proposed uniform inequality asks whether, for every $\varepsilon > 0$, there exist integers N and R such that

$$M_{m,n}(\theta) \le c_\theta + \varepsilon \quad \text{whenever } n \ge N \text{ and } m/n \ge R,$$

and whether no smaller constant has this property. The supremum and quantifiers specify the source's uniform $1 + o(1)$ bound; the floor convention handles row-count integrality.

Connection to numerical linear algebra

This asks how well any unit-row linear system can retain its smallest singular value under worst-case row removal, a geometric constraint on robust iterative solvers.

References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, the third question immediately after equation (8). Published paper. Author preprint, v1, §2.3, paragraph following (⋄).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1 (background on the same restricted-singular-value heuristic and the different streaming model), §5. Preprint.

Status check

Checked the published question and latest arXiv records on 2026-09-08. Targeted searches for "Steinerberger subsingular", "small subsingular values", "quantile Kaczmarz optimal constants", "unit rows smallest singular value", and 2025/2026 variants found no resolution of the all-matrices extremum. Recent streaming Kaczmarz guarantees do not supply this universal extremal inequality. The random-ensemble limit in the same source is a related but distinct question: it specifies a random construction, whereas this question optimizes over all unit-row matrices. Openness remains subject to the limits of this search.